the work done is given by $dW = -E\,dx$. The minus sign implies that work is done against the electric field. This work done is equal to electric potential difference. Therefore,

$$dW = dV.$$

(or) $$dV = -E\,dx \qquad (1.42)$$

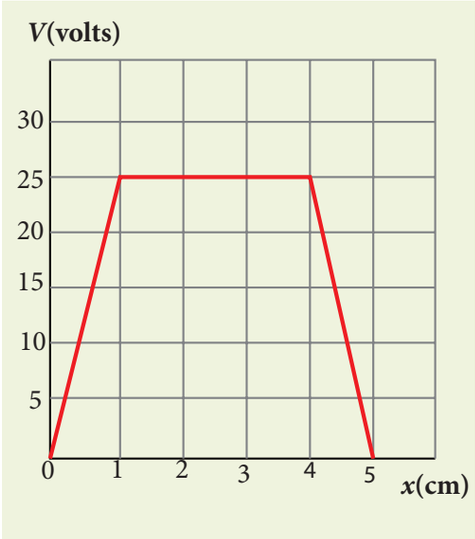
Hence $$E = -\frac{dV}{dx} \qquad (1.43)$$

The electric field is the negative gradient of the electric potential. In general,

$$\vec{E} = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial k}\hat{k}\right) \qquad (1.44)$$

EXAMPLE 1.14

The following figure represents the electric potential as a function of x – coordinate. Plot the corresponding electric field as a function of x.



Solution

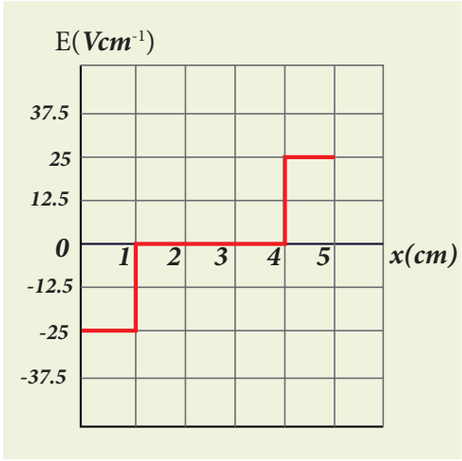
In the given problem, since the potential depends only on x, we can use $\vec{E} = -\frac{dV}{dx}\hat{i}$ (the other two terms $\frac{\partial V}{\partial y}$ and $\frac{\partial V}{\partial z}$ are zero)

From 0 to 1 cm, the slope is constant and so $\frac{dV}{dx} = 25\,V\,cm^{-1}$. So $\vec{E} = -25\,V\,cm^{-1}\hat{i}$

From 1 to 4 cm, the potential is constant, V = 25 V. It implies that $\frac{dV}{dx} = 0$. So $\vec{E} = 0$

From 4 to 5 cm, the slope $\frac{dV}{dx} = -25\,V\,cm^{-1}$. So $\vec{E} = +25\,V\,cm^{-1}\hat{i}$.

The plot of electric field for the various points along the *x* axis is given below.



1.5.6 Electrostatic potential energy for collection of point charges

The electric potential at a point at a distance *r* from point charge q_1 is given by

$$V = \frac{1}{4\pi\varepsilon_\circ}\frac{q_1}{r}$$

This potential *V* is the work done to bring a unit positive charge from infinity to the point. Now if the charge q_2 is brought from infinity to that point at a distance *r* from q_1, the work done is the product of q_2 and the electric potential at that point. Thus we have

$$W = q_2 V$$

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